

Buddy: Helping School-age Children to Combat Bullying

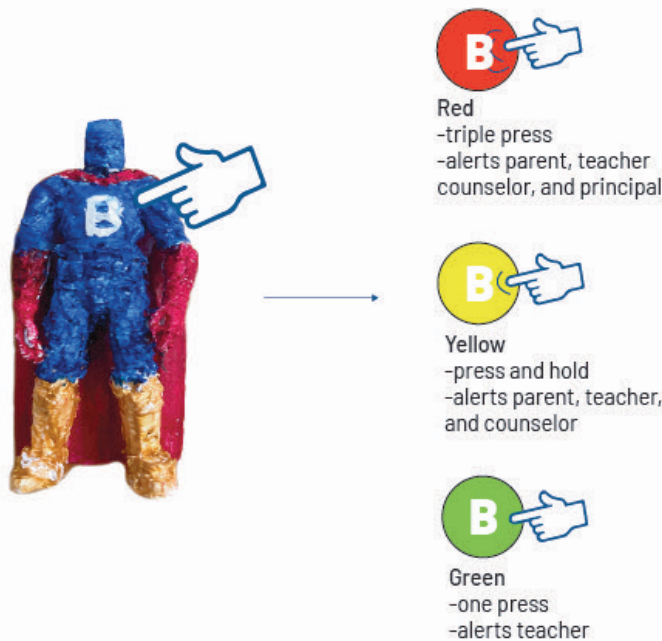
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Buddy Key Chain



Buddy App



Figure 1: An overview of the *Buddy* Key Chain and the *Buddy* App. When a student is bullied or witnesses bullying, he can press the button on the Key Chain and send an alert to the app. Different notifications indicate different levels of risk. The student's name, location, and time of the bullying incident will be displayed on the home page of the app.

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ABSTRACT

Bullying has been an ongoing and prevalent problem at school for years. School-age children need a way to feel safe and protected in their own school and be able to ask for help when being bullied. With the goal of improving quality education, we created the *Buddy* key chain and app aiming at combating bullying in regard to school-age children, parents, and teachers involved in the issue of bullying in elementary schools and middle schools. The *Buddy* key chain is an automatic, discreet device with a secret button that alerts parents and teachers of instances of bullying directly from the

student who is bullied or witnesses bullying. When the button is pressed, the parents and teachers attached to that student's account will be alerted. This provides the student with a way to report the bullying without encouraging the bullying to continue. The issue this device solves is the lack of quality education. Students cannot learn effectively if they do not feel safe in their learning environment.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design**; • **Interaction design process and methods** → *User interface design*.

KEYWORDS

Quality education, bully, interaction design

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1 INTRODUCTION

Bullying is a persistent phenomenon that is extremely prevalent at school. Around 20 percent of American children have been bullied in elementary school, and about 20 percent of American children have described themselves as having engaged in bullying behavior [1]. Bullying, which can also be described as aggressive relations, ranges from teasing people's names to a group of students physically abusing a classmate. We should focus on bringing awareness to the bullying problem and tackling the issue as soon as it arises.

Students are not the only point of action. Parents, teachers, counselors, and other school faculty should get involved in the situation and take action. Many studies prove that bullying occurs when there is an audience [4]. Bullying is normally left unresolved at school unless a student or bystander speaks up or a teacher or staff member notices. Some busy parents struggle to communicate with teachers about their children. Whether they are working or not, there is usually not a lot of conversation between the two. When some students get into trouble, they may feel embarrassed or unwilling to ask for help. Students that are uncomfortable with speaking up or need assistance doing so need a way to get help when they are in an uncomfortable situation.

This project contributed to the quality education goal of the 17 Sustainable Development Goals identified by the United Nations [3]. With the goal of improving the widely experienced issue of bullying between young students in schools and increasing student safety, we designed *Buddy*, a smart key chain with an app to alert parents and teachers about bullying and provide a safe and convenient way for school-age children to ask for help (Figure 1). The *Buddy* key chain has an alert button that is connected to an app that parents, teachers, and counselors can use. When a student presses the button, teachers and/or counselors that have the app and student info within the app will receive a notification on the *Buddy* that displays when and where the bullying instance occurs. Parents will then receive a notification that their child has pressed the button. Furthermore, the *Buddy* app provides an easy way for

teachers and parents to communicate and follow up with the Bullying instance. We aim to create a solution that helps teachers and parents participate better and engage more in the mitigation of aggression relations between students.

2 RELATED WORK

There are apps similar to *Buddy* that focus on tracking the child or warning signs of bullying. For example, an app called *Bully Button*, allows the child to report bullying by recording a video and sending it to the parents and the school authority. According to many school rules and regulations, it is not practical for a child to pull their phone/tablet out to record a video during school, which is why the key chain idea for our project will act as a better solution.

Additionally, The app *KnowBullying* [5] in conjunction with Stop-Bullying.gov [6] helps children, parents, and educators recognize the warning signs of bullying and also gives parents and educators tips about how bullying is affecting their children based on their specific age group. *Security Bubble* [9], a bracelet that children can wear, tracks their location even in areas that don't have cellular reception. *Ripple* [7] is another small and inconspicuous safety device that can be clipped onto things like a necklace, purse, or lanyard that you click to call 911 in case of emergency. We also found that wearable emergency buttons have been widely used in elderly care [2, 8].

A common issue within the current apps is that they are aimed at more parental and educator use, and are not necessary for a young child to use this app. Additionally, there was nothing made to look discreet for a child to use in situations where calling for help may worsen the situation. Our goal for *Buddy* was to make schools feel safer for students. Bullying within schools hinders students from receiving a safe, quality education, but with the use of *Buddy*, students will feel safer communicating with authority about instances of bullying.

3 INITIAL USER RESEARCH

To understand student, parent, and teacher perspectives on bullying, we conducted 12 semi-structured interviews with five students (three female, two male) between the ages of eight and 14 years old, three parents (one mother and two fathers), two teachers and two counselors between the ages of 22 and 53 years old. We recruited participants from public elementary and middle schools in the U.S. The interviews were conducted either in person or remotely via Zoom. The overarching interview questions included students' witnesses or experiences with bullying at school, parents' attitudes toward bullying, and teachers' strategies for handling bullying in school.

We used affinity diagramming to synthesize insights and identified four themes from the interviews:

- (1) Most of the students trust their teachers and staff to help them in a situation regarding bullying. We also noticed from our teachers that they used positive reinforcement. If that did not work, the counselor would take care of the problem from there. For example, one of the middle school educators we interviewed stated:

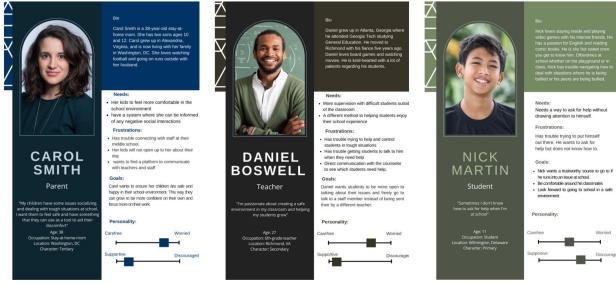


Figure 2: From left to right: the parent persona, the teacher persona, and the student persona.

"I usually deal with instances of bullying in my classroom on my own, but if it gets too big of a situation, I trust the counselors to handle the situation."

- (2) Teachers struggled with handling students and their behaviors when there was less supervision. An example from a teacher we interviewed was when she was helping another student in the classroom, she cannot have her eyes on all of her kids. Oftentimes, parents and teachers won't know about instances of bullying unless the child or a bystander speaks up about the situation. Other cases included on the playground or in the lunchroom. A father of an elementary school student said:

"It is nice to hear about what happens throughout the school day from someone other than my child, especially if there are instances of bullying."

- (3) We observed students did trust their teachers/counselors, however, they were unsure of how to go about asking or talking to them when they most needed it. In this case, if a teacher was aware of the situation, they sent them to the counselor for more help. Moreover, we found that there was always an inconsistent form of communication between teachers and the administration. An administrator at an elementary school stated:

"Sometimes details of situations regarding bullying get lost in communication and it is sometimes difficult to discuss matters regarding the bullying of students."

To empathize with users, we developed three personas (Figure 2) to illustrate users' end goals, needs, and frustrations, as well as three user journey maps to visualize their experience with bullying.

4 IMPLICATIONS FOR DESIGN

From the results of our initial user research, we draw the following requirements for designing a convenient and effective tool to combat bullying at school.

- (1) When interviewing parents, we found that many of them wanted easy communication with their children at school, especially in situations of bullying. The *Buddy* device and the app should allow parents to be aware if their child is

being bullied at school and give them more insight into the situation. The *Buddy* app should enable parents to communicate with administration and teachers regarding bullying in an easier fashion.

- (2) The key chain should give students a way to call for help without drawing attention to the situation. It allows them to call for help when they feel too scared or embarrassed to bring attention to the bullying.
- (3) When interviewing administration, we found that when they have to deal with instances of bullying, sometimes there was little communication between the teacher and counselor regarding the instance of bullying. The *Buddy* app should give a way for teachers to communicate the bullying issue in a convenient way.

5 DESIGN PROCESS

5.1 Ideation

To achieve these goals, we created dozens of iterations and brainstormed ideas including self-esteem exercises implemented into the daily class lessons, an interactive video game that gives real-life examples of bullying, and a daily check-in on an iPad when students come into class. After getting feedback from students, we concluded that they should have a way of being able to tackle the problem themselves. Practices like holding assemblies to bring awareness on bullying are not the most effective way for students to truly absorb the importance of this issue.

After several rounds of discussion, our team thought about designing a wearable device like a necklace or wristband that allows students to press buttons on the device to alert teachers when being bullied or witnessing acts of bullying. This was not practical, as it was too obvious for other students to see. We wanted to create something more discrete. Therefore, we sketched ideas of physical key chains, which would fit onto the backpacks of students. We decided to have one singular button to alert the counselor when it was pressed. One button would be the most user-friendly for young kids, and having multiple buttons could be more confusing. We decided the best way to have a discrete key chain would be to make it look like an action figure of popular characters (e.g., Disney and Marvel).

We walked through our idea with students, parents, and teachers and received positive feedback from them: 1) Students mentioned the key chain could provide a direct source of help at the bullying moment, and 2) Parents emphasized that they wanted an easier form of communication with the school. Therefore, we decided to go with the *Buddy* key chain for students to use and the *Buddy* app for parents and teachers.

5.2 Initial Design

We started our initial design with a low-fidelity prototype and tested it with users. The low-fidelity prototype of the *Buddy* Key chain gave students an option of what they wanted it to look like, so it could be discrete from others. The *Buddy* app consisted of a messaging system for teachers and parents as well as a files page for children's medical information and emergency contact. We have tasks showing a teacher getting in contact with a parent through

the files function and the teacher getting a notification from the app, causing them to enter the app to message a parent.

5.3 User Testing and Iterations

We conducted user testing with a think-aloud protocol with five students and five teachers to gain feedback on the *Buddy* low-fidelity prototype. Participants were asked to perform six tasks and verbalize their thoughts as they moved through the interface. Then we asked a list of follow-up questions based on explicit behaviors that we observed.

We received a lot of positive feedback from our participants, along with constructive suggestions on what to fix. The messages tab for communication between teachers and parents was greatly appreciated, making the process quick and easy when a child was in danger. Most of the participants stated they found this aspect easy to use and would be more efficient than emailing the school. Moreover, some participants mentioned that they wanted the app to track where and when the *Buddy* button was pressed. Participants expected to see a more convenient and easy way to pair the key chain with the app and receive notifications directly from the key chain.

To reinforce these principles in our design, we made three improvements:

- (1) Change the homepage to display unread messages and alerts. Allow users to track when and where the *Buddy* button was pressed and who pressed it.
- (2) Allow users to manage appointments, *Buddy* notifications, and medical information all under the files tab.
- (3) Include a device settings tab so parents can pair the key chain and locate their children when the key chain button is pressed. This ensures Children's safety and allows parents to check on their children at the moment.

6 FINAL DESIGN

Our final design includes six task flows with upgraded changes to the layout and technicalities of the app, creating a better overall aesthetic and user experience. This section describes the final design of the *Buddy* key chain and the app.

6.1 System Design

Each *Buddy* key chain consists of a microcontroller, GSM module, GPS module, solar panel for charging purposes, and voice playback circuit. Information between the Key Chain and the app is transmitted and received via Wi-Fi or as GSM data. The key chain can be paired with the app using network connections. Each key chain has a unique barcode located at the bottom of the key chain. Scanning the barcode will pair the key chain with the app. The key chain has a switch on the foot of the action figure/character to connect to Bluetooth. Additionally, when the key chain is located in connectivity-constrained areas, GSM technology provides communication capability to transmit data to the app. The child's position is tracked by GPS and sent to the microcontroller, which can forward the GPS data to the GSM board. In turn, GSM sends the corresponding child's GPS location to the parent's app. The process of this is anonymous to protect the privacy of the child and the owner of the device with the *Buddy* app. The key chain has a

solar panel to charge while the student walks to school or plays at recess.

Moreover, the *Buddy* app allows the child to record their voice in an emergency situation. In a complex situation such as a physical altercation, when the child yells "help, help, help" or screams, it triggers the voice playback circuit and sends a text message to their parents' phones as well as alerts parents on the app.

The *Buddy* key chain offers various design options and customizable choices for a child to choose from. Categories like age-appropriate action figures or their favorite television characters will provide a stronger bond between the student and the *Buddy* key chain. Another option offered is characters with soft and fuzzy textures, for example, Winnie the Pooh and Sully from *Monsters Inc.*

6.2 Send Alerts From the *Buddy* Key Chain to the App

When a student is bullied or witnesses someone being bullied, they can press the button one time on the key chain and send an alert to the *Buddy* app. Teachers will get a green notification that a student used the *Buddy* button. The location and time of the bully incident will be displayed on the home page. Teachers can click on the notification to look further into what happened and message the parent to set up an appointment to meet. If the student holds down the *Buddy* button, this sends a yellow alert notification, meaning that the student needs more immediate help. This notification will go to the student's teacher, parent, and counselor. If the situation is more of an emergency, the student can triple-click the *Buddy* key chain, or they can yell "help, help, help" or scream, which will send an alert to all parties involved including parents, teachers, counselors, and principals (see Figure 3).

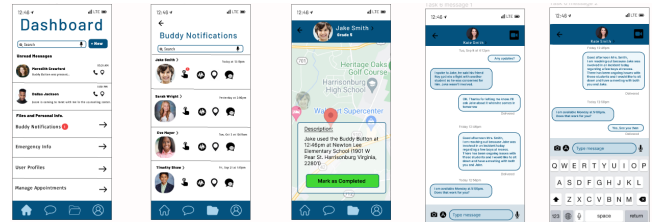


Figure 3: A student presses the button on the *Buddy* key chain and his teacher is alerted through the app

6.3 Communicate with Parents and Schedule Appointments

If a parent does not respond to a teacher about their child bullying other students, the teacher can reach out through files to try and get in contact with the parent (Figure 4). This could be a common issue within the school system, so we wanted to include a solution. The teacher is able to locate and message parents directly within the "Emergency Information" tab. Users can message each other in order to set up an appointment. There is also a "Manage Appointments" button on the dashboard page for all users, where appointments can be scheduled efficiently. The "Manage Appointments" feature is also where users can find their appointment history.

6.4 Communicate with Teachers and Students meeting with Counselor

When the student presses the *Buddy* button, the teacher and counselor are alerted. The teacher can communicate with the counselor about the solution and meet with the student. They can message each other in order to set up an appointment. There is also a "Manage Appointments" button on the dashboard for all users, where appointments can be scheduled efficiently. The "Manage Appointments" feature is also where users can find their appointment history.

6.5 Privacy and Safety Considerations

Buddy ensures the safety of students with communication between parents and counselors and limited access to students' current locations. Once students leave their school grounds (about a one-mile from the campus grounds), their location will no longer be visible to teachers and counselors. The parent will still have access to their child's location.

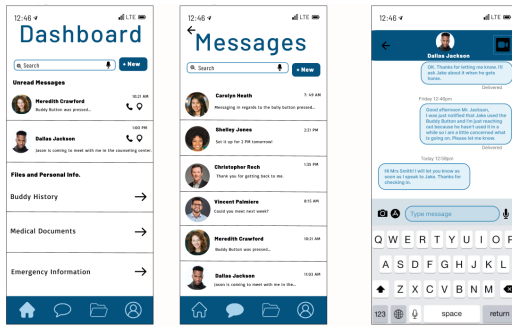


Figure 4: A teacher is getting in contact with a counselor (Dallas Jackson). They want to message a student.

7 EVALUATION AND FUTURE WORK

After modifying our high-fidelity prototype to its optimal potential, we conducted user testing with 12 participants. Participants were asked to complete six tasks ranging from using the key chain to call for help, creating an account, messaging a parent, and inputting profile information. We concluded with a SUS score of approximately 75, which means this prototype is in good condition to continue to develop in the future. The feedback we received from children interacting with the key chain was positive. Children found the key chain easy to use and understood how to interact with it in different situations. The biggest issue was getting situated where the different features were on the app. The dashboard on the home page was extremely useful for this reason.

In the future, we will continue to develop this app with features that are up to date with current technological features. Through more user research and having the app be used in schools with real students, parents, and educators, we will update the app and its features based on the user's needs.

8 CONCLUSIONS

This study aimed to find a way for students to report bullying while feeling safe and provide a safer environment for them while at school. *Buddy* encourages students to ask for help, and know that there are people there to support and help them. A limitation we found would be getting the device and app implemented into a school and the school's routine. Additionally, getting *Buddy* approved to be used in schools would be a limitation. Suggestions for future research would be to have a trial run of the device and app in a school setting.

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